

## 5.5 Surface Integrals\_Guide

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5.5 Surface  
Integrals\_...

## 5.5 Surface Integrals

**Motivation:**

In section 5.2.1 we discussed the idea that a line integral was useful for adding up a quantity over a given curve  $C$ . In this section we wish to extend this idea to adding up a quantity over a given *surface*  $\sigma$  to create the idea of a surface integral.

This idea can be useful in a variety of applications

- Adding up charge across a magnetized surface.
- Adding up gravitational forces across a surface.
- Adding up the density across a surface to help find the mass of the surface.
- Adding up flow of fluid through  $\sigma$  to help find the total quantity of fluid flowing through  $\sigma$ .

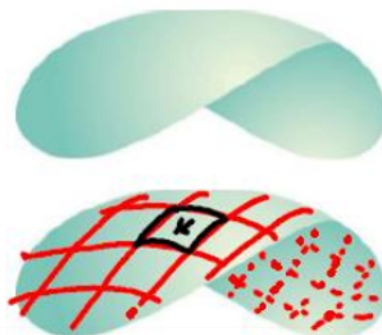
So, how do we construct an integral that will do this?

**Exploration:** Suppose we have a smooth surface  $\sigma$  that we plan to move across, and we want to add up the quantity at each point given by the function  $f(x, y, z)$ . Let  $Q$  represent the total quantity over the surface. How do we find  $Q$ ?

We can begin by partition  $\sigma$  up into  $n$  patches.

Consider the  $k^{\text{th}}$  patch and choose a representative point

$$P_k(x_k^*, y_k^*, z_k^*)$$



Let  $\Delta S_k$  represent the surface area of the  $k^{\text{th}}$  patch,

&  $Q_k$  represent the quantity that exists over this patch.

We will let  $f(x_k^*, y_k^*, z_k^*)$  be an estimation of the quantity we want over this patch.

Then we can say  $Q_k \approx f(x_k^*, y_k^*, z_k^*) \cdot \Delta S_k$

$$\text{So } Q \approx \sum_{k=1}^n Q_k = \sum_{k=1}^n f(x_k^*, y_k^*, z_k^*) \cdot \Delta S_k$$

$$\text{Therefore, } Q = \lim_{n \rightarrow \infty} \left[ \sum_{k=1}^n f(x_k^*, y_k^*, z_k^*) \cdot \Delta S_k \right] = \iint_{\sigma} f(x, y, z) \, ds$$

This leads us to the following definition.



**Definition:** Assuming that  $f$  is continuous, the surface integral of  $f$  over the surface  $\sigma$  is  $\iint_{\sigma} f(x, y, z) dS$ .

**Remark:** Let  $\mathbf{r}(u, v) = x(u, v)\mathbf{i} + y(u, v)\mathbf{j} + z(u, v)\mathbf{k}$  be the parameterization of the surface  $\sigma$ . We saw in section 4.4 that the surface area of the  $k^{\text{th}}$  patch,  $\Delta S_k$ , is approximated by  $\Delta S_k \approx \|\mathbf{r}_u \times \mathbf{r}_v\| dA$ .

Therefore, we can rewrite the previous definition as follows.

**Definition (Updated):** Assuming that  $f$  is continuous, the surface integral of  $f$  over the surface  $\sigma$  is

$$S = \iint_{\sigma} f(x, y, z) dS = \iint_D f(\mathbf{r}(u, v)) \|\mathbf{r}_u \times \mathbf{r}_v\| dA_{uv}$$

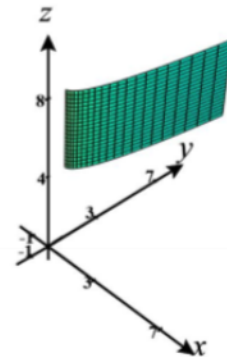
**Remark:** Note that if  $f(x, y, z) = 1$ , then the surface integral  $\iint_{\sigma} f(x, y, z) dS$  becomes  $\iint_{\sigma} dS = \iint_D \|\mathbf{r}_u \times \mathbf{r}_v\| dA$  which we know from section 4.4 yields the surface area of  $\sigma$ .

**Example 1:** Consider the surface  $\sigma = \{(x, y, z) \mid \overset{\text{function}}{\downarrow} y = 2x^2 + 1, 0 \leq x \leq 2, 4 \leq z \leq 8\}$ .

a) Find a parameterization  $\mathbf{r}(u, v)$  for  $\sigma$ .

$$\text{let } x = u \quad y = 2u^2 + 1 \quad z = v$$

$$\text{So } \mathbf{r}(u, v) = \left\langle \underset{x}{u}, \underset{y}{2u^2 + 1}, \underset{z}{v} \right\rangle \text{ for } 0 \leq u \leq 2, 4 \leq v \leq 8$$



b) Evaluate  $\iint_{\sigma} (xz^2) dS$ .

$$\text{Note that } \mathbf{r}_u \times \mathbf{r}_v = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 1 & 4u & 0 \\ 0 & 0 & 1 \end{vmatrix} = \langle 4u, -1, 0 \rangle$$

$$\text{So } \|\mathbf{r}_u \times \mathbf{r}_v\| = \sqrt{16u^2 + 1}$$

$$\text{Therefore } \iint_{\sigma} (xz^2) dS = \int_0^2 \int_4^8 (uv^2) \sqrt{16u^2 + 1} du dv = \dots = \frac{28}{9} (65^{3/2} - 1)$$

**Example 2:** Compute the surface integral  $\iint_{\sigma} z^2 dS$  where  $\sigma$  is the unit sphere  $x^2 + y^2 + z^2 = 1$ .

Hint: Use what you know about spherical coordinates to find parametric equations for  $\sigma$ .

Recall from spherical coordinates that

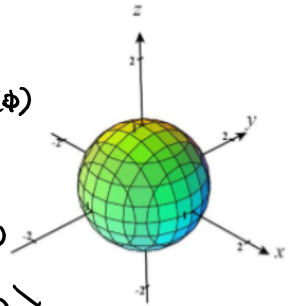
$$x = \rho \sin(\phi) \cos(\theta) \quad y = \rho \sin(\phi) \sin(\theta) \quad z = \rho \cos(\phi)$$

Here  $\rho = 1$ . Let  $u = \phi$  &  $v = \theta$ .

$$\text{Then } x = \sin(u) \cos(v) \quad y = \sin(u) \sin(v) \quad z = \cos(u)$$

$$\text{So } \vec{r}(u, v) = \left( \underbrace{\sin(u)}_x \underbrace{\cos(v)}_y, \underbrace{\sin(u)}_y \underbrace{\sin(v)}_z, \underbrace{\cos(u)}_z \right)$$

$$0 \leq u \leq \pi \quad \& \quad 0 \leq v \leq 2\pi$$



$$\text{Note } \vec{r}_u \times \vec{r}_v = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \cos(u) \cos(v) & \cos(u) \sin(v) & -\sin(u) \\ -\sin(u) \sin(v) & \sin(u) \cos(v) & 0 \end{vmatrix}$$

$$= \langle \sin^2(u) \cos(v), \sin^2(u) \sin(v), \sin(u) \cos(u) \cos^2(v) + \sin(u) \cos(u) \sin^2(v) \rangle$$

$$= \langle \sin^2(u) \cos(v), \sin^2(u) \sin(v), \sin(u) \cos(u) \rangle$$

$$\text{So } \|\vec{r}_u \times \vec{r}_v\| = \sqrt{\sin^4(u) \cos^2(v) + \sin^4(u) \sin^2(v) + \sin^2(u) \cos^2(u)} \quad \text{b/c } 0 \leq u \leq \pi$$

$$= \sqrt{\sin^4(u) + \sin^2(u) \cos^2(u)} = \sqrt{\sin^2(u)} = |\sin(u)| \stackrel{\downarrow}{=} \sin(u)$$

$$\text{Therefore } \iint_{\sigma} z^2 dS = \int_0^{2\pi} \int_0^{\pi} \cos^2(u) \sin(u) du dv = \dots = \frac{4\pi}{3}$$

**Example 3:** Suppose  $\sigma$  is a smooth surface given by  $z = g(x, y)$  where  $(x, y) \in D$ . Express the surface integral  $\iint_{\sigma} f(x, y, z) dS$  as a double integral in terms of  $x$  and  $y$ .

We first parameterize  $\sigma$ . Let  $x = u$ ,  $y = v$ ,  $z = g(u, v)$

$$\vec{r}(u, v) = \langle u, v, g(u, v) \rangle$$

$$\text{So } \vec{r}_u \times \vec{r}_v = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 1 & 0 & g_u(u, v) \\ 0 & 1 & g_v(u, v) \end{vmatrix} = \langle -g_u(u, v), -g_v(u, v), 1 \rangle$$

$$\text{Thus } \|\vec{r}_u \times \vec{r}_v\| = \sqrt{g_u(u, v)^2 + g_v(u, v)^2 + 1}$$

$$\begin{aligned} \text{So } \iint_{\sigma} f(x, y, z) dS &= \iint_{D_{uv}} f(u, v, g(u, v)) \sqrt{g_u(u, v)^2 + g_v(u, v)^2 + 1} dA_{uv} \\ &= \iint_{D_{xy}} f(x, y, g(x, y)) \sqrt{g_x(x, y)^2 + g_y(x, y)^2 + 1} dA_{xy} \end{aligned}$$

*Note: In this example we are assuming that  $f$  is continuous on  $\sigma$ . Oftentimes this condition is just assumed without being stated in the textbook.*

The most recent example is the justification for the following formulas.

**Formulas:** Suppose  $\sigma$  is a smooth surface given by  $z = g(x, y)$  where  $(x, y) \in D$ . Then,

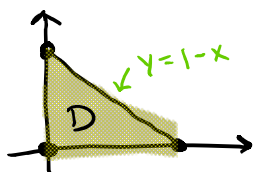
1. A normal vector to the surface at  $(x, y, g(x, y))$  is  $\langle -g_x(x, y), -g_y(x, y), 1 \rangle = \vec{r}_u \times \vec{r}_v$

$$2. \iint_{\sigma} f(x, y, z) dS = \iint_D \left[ f(x, y, g(x, y)) \sqrt{\left(\frac{\partial z}{\partial x}\right)^2 + \left(\frac{\partial z}{\partial y}\right)^2 + 1} \right] dA$$

*Note: It is possible to derive similar formulas in the cases where the surface is given by  $y = h(x, z)$  or when the surface is given by  $x = j(y, z)$ . Can you guess what the formulas would be in these cases? See page 1133 of the book for one of the formulas.*

**Example 4:** Fully setup, but do not evaluate, the surface integral  $\iint_{\sigma} 2 \cos(y)(z - \tan(y) + 1) dS$  where  $\sigma$  is the portion of the surface  $z = \tan(y)$  which exists over the triangular region with endpoints  $(0,0)$ ,  $(1,0)$ , &  $(0,1)$ .

$$\begin{aligned} \iint_{\sigma} 2 \cos(y) (z - \tan(y) + 1) dS &= \iint_D 2 \cos(y) (\tan(y) - \tan(y) + 1) \sqrt{0 + \sec^4(y) + 1} dA \\ &= \iint_D 2 \cos(y) \sqrt{\sec^4(y) + 1} dA \end{aligned}$$



$$D = \{(x, y) \mid 0 \leq y \leq 1 - x \text{ \& } 0 \leq x \leq 1\}$$

$$\text{So we get } \int_0^1 \int_0^{1-x} 2 \cos(y) \sqrt{\sec^4(y) + 1} dy dx$$

We now state a fact which we will recognize as an extension of a property of line integrals.

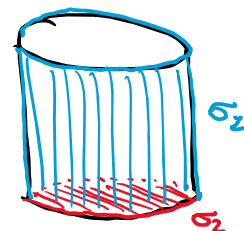
**Fact:** If  $\sigma$  is a finite union of smooth surfaces:  $\sigma_1, \sigma_2, \dots, \sigma_n$  (we say  $\sigma$  is piecewise-smooth), then

$$\iint_{\sigma} f(x, y, z) dS = \iint_{\sigma_1} f(x, y, z) dS + \iint_{\sigma_2} f(x, y, z) dS + \dots + \iint_{\sigma_n} f(x, y, z) dS.$$

**Example 5:** Suppose that  $\sigma$  is the “can without a top” which consists of  $\sigma_1 = \{(x, y, z) \mid x^2 + y^2 = 1, 0 \leq z \leq 2\}$  and  $\sigma_2 = \{(x, y, z) \mid x^2 + y^2 \leq 1, z = 0\}$ .

a) Find a parameterization  $\mathbf{r}(u, v)$  of  $\sigma_1$ .

$$\bullet \vec{r}_1(u, v) = \left\langle \frac{\cos(u)}{x}, \frac{\sin(u)}{y}, \frac{v}{z} \right\rangle \quad \begin{matrix} 0 \leq u \leq 2\pi \\ 0 \leq v \leq 2 \end{matrix}$$



b) Find a parameterization  $\vec{r}_2(u, v)$  of  $\sigma_2$ .

$$\bullet \vec{r}_2(u, v) = \left\langle \frac{v \cos(u)}{x}, \frac{v \sin(u)}{y}, \frac{0}{z} \right\rangle \quad \begin{matrix} 0 \leq u \leq 2\pi \\ 0 \leq v \leq 1 \end{matrix}$$



c) Evaluate  $\iint_{\sigma} (x^2 + y^2 + z^2) dS$ .

$$\iint_{\sigma} (x^2 + y^2 + z^2) dS = \underbrace{\iint_{\sigma_1} (x^2 + y^2 + z^2) dS}_{\text{blue}} + \underbrace{\iint_{\sigma_2} (x^2 + y^2 + z^2) dS}_{\text{red}}$$

Note  $\vec{r}_u \times \vec{r}_v = \langle \cos(u), \sin(u), 0 \rangle \rightarrow \|\vec{r}_u \times \vec{r}_v\| = 1$

$\vec{r}_u \times \vec{r}_v = \langle 0, 0, -v \rangle \rightarrow \|\vec{r}_u \times \vec{r}_v\| = v$

$$= \underbrace{\int_0^2 \int_0^{2\pi} (1 + v^2) \cdot 1 \, du \, dv}_{\text{blue}} + \underbrace{\int_0^1 \int_0^{2\pi} v^2 \cdot v \, du \, dv}_{\text{red}}$$

d) Note that  $\sigma_2$  didn't need to be parameterized and could have just been expressed as the surface in the form of  $z = f(x, y)$  over a region  $D$ . Find the surface integral for  $\sigma_2$  using this approach.

$\sigma_2$  is given by  $z = 0$  over a region  $D = \{(r, \theta) \mid 0 \leq r \leq 1 \text{ \& } 0 \leq \theta \leq 2\pi\}$

$$\begin{aligned} \iint_{\sigma_2} (x^2 + y^2 + z^2) dS &= \iint_D (x^2 + y^2 + 0) \sqrt{0^2 + 0^2 + 1} \, dA \\ &= \underbrace{\int_0^{2\pi} \int_0^1 (r^2) r \, dr \, d\theta}_{\text{red}} \end{aligned}$$